

PANEL DATA FROM REPEATED CROSS-SECTIONS

Pseudo-Panels

Class 3 · Hour 1

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Panel Data · EBAPE/FGV

CLASS 3 · HOUR 1

What We'll Cover This Hour

01 **The problem pseudo-panels solve**
No true panel, but a series of independent cross-sections

02 **Cohorts: what they are and why they work**
Tracking groups instead of individuals across surveys

03 **From individual to cohort equations**
How Deaton (1985) builds a panel model out of cross-sections

04 **The errors-in-variables problem**
Sample cohort means are consistent, but not error-free

05 **Design choices and Stata**
Cohort size, identification, and a hands-on example

The Problem: Real Panels Are Often Missing

> *"In many countries, there are few or no panel data, but there exists a series of independent cross-sections."*
> — Deaton (1985)

1	**Deaton's example**	The UK had no panel data on consumer expenditure or household labor supply — but large household surveys ran every year, each with a fresh, independent sample
2	**The catch**	Samples are drawn anew each year, so individual households cannot be followed over time
3	**Deaton's question**	Can we still recover panel-like relationships — including fixed effects — from these data?

Familiar from Class 1: this is a **repeated cross-section**, not a panel. Today we turn one into the other.

Why This Matters: Cross-Sections vs. Time Series

Deaton's motivating puzzle: coefficients estimated from cross-sections often differ substantially from those estimated with time series.

The puzzle

Documented at least since Kuznets: savings ratios rise more with income in cross-sections than in aggregate time series; expenditure elasticities are more dispersed in cross-sections.

The diagnosis

Inadequate statistical control. Poor and rich households differ in more than income, and a long list of demographics cannot fully fix that.

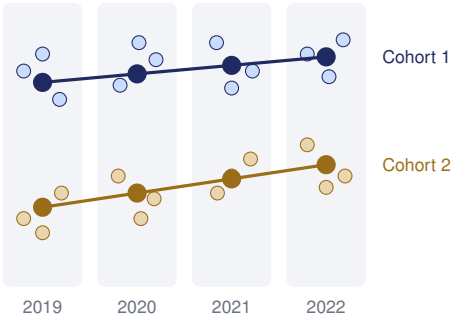
The natural fix

Individual fixed effects — each household as its own control. But that requires a true panel. . .

. . . unless we can build something that behaves like one.

The Core Idea: Track Cohorts, Not Individuals

Individuals change from survey to survey, but groups with **fixed membership** — cohorts — reappear in every survey.



- Small circles: **different people** each year
- Large dots: the cohort's **mean** in that survey
- Each cohort's means form a time series: the cohort becomes the **panel unit**

What Counts as a Cohort?

The most obvious example is an age cohort, but Deaton stresses the idea is far more general.

Birth cohorts

All men born between 1945 and 1950, followed as they age through each survey

Historical cohorts

Vietnam War veterans: a group defined once, and identifiable indefinitely

Institutional cohorts

Graduates of a given program and year (say, the EBAPE PhD class of 2015)

The requirement: membership must be fixed and observable in every survey. Traits people can change — region of residence, occupation, income bracket — make the cohort's composition drift over time.

The Individual Model, With a Fixed Effect

In the notation of this course (household h, period t):

$$y_{ht} = \beta x_{ht} + \theta_h + \varepsilon_{ht}$$

1	**θ_h: the household's own fixed effect**	Uses each household as its own control — exactly what a real panel offers
2	**θ_h is generally correlated with x_{ht}**	So the equation can only be estimated consistently with FE — which needs the same h observed repeatedly
3	**With repeated cross-sections, each h appears once**	Unit FE is infeasible . . . unless h belongs to a well-defined, trackable cohort

Deaton's application is an Engel curve: y is a good's budget share, x is log total outlay, plus demographics. In his paper the index i denotes *goods*, not units.

Why Averaging Within Cohorts Works

Deaton's key requirement: the individual relationship must be **linear in parameters** (not necessarily in variables). Average both sides over *all* population members *h* of cohort *c* at time *t* (starred = population means):

$$y^*_{ct} = \beta\, x^*_{ct} + \theta^*_c + \varepsilon^*_{ct}$$

1	**Averaging preserves the linear structure**	The parameters carry through unchanged
2	**Household FE become cohort FE**	Membership is fixed, so θ^*_c is a genuine constant — estimable with cohort dummies, since each cohort reappears every period
3	**The catch**	We never observe the starred population means — only *sample* cohort means from each survey

The Key Insight: Consistent but Error-Ridden

"The sample cohort means from the surveys are consistent but error-ridden estimates of the true cohort means." — Deaton (1985)

$$\bar{x}_{ct} = x^*_{ct} + \underbrace{(\bar{x}_{ct} - x^*_{ct})}_{\text{sampling error}}, \qquad \text{Var}(\bar{x}_{ct} - x^*_{ct}) \approx \frac{\sigma^2_{x,ct}}{n_{ct}}$$

1. **The error shrinks as n_{ct} grows** — Consistent — but with the finite cells of any real survey, every mean carries sampling error
2. **A classic errors-in-variables problem** — Fixable, because the error variance can be estimated from the micro data

Why You Can’t Just Run OLS on Cohort Means

Every variable — not just y — is a sample mean, measured with sampling error.

Classical case: error in y only

OLS remains consistent: independent measurement error in the dependent variable is absorbed into the residual.

Pseudo-panel: error in x too

Sample means of x are error-ridden: errors-in-variables, which attenuates $\hat{\beta}$ (toward zero, with one regressor).

And the errors in $\bar{y}_{ct}$ and $\bar{x}_{ct}$ are **correlated** — they come from the same sampled individuals.

The FE (within-cohort) transformation makes this worse: it removes signal (θ_c^*) but keeps all the noise.

The Fix: Errors-in-Variables Correction

Deaton uses the micro survey data twice: once for the cohort means, once for their sampling (co)variances.

$$\tilde{\boldsymbol{\beta}} = \left(\boldsymbol{M}_{xx} - \boldsymbol{\Sigma}\right)^{-1}\left(\boldsymbol{m}_{xy} - \boldsymbol{\sigma}\right)$$

1	**$\boldsymbol{M}_{xx}$ and $\boldsymbol{m}_{xy}$: ordinary sample moments**	Computed from the observed cohort means (within-cohort deviations under FE), exactly as in OLS
2	**$\boldsymbol{\Sigma}$ and $\boldsymbol{\sigma}$: sampling (co)variances**	Of the errors in $\bar{x}_{ct}$, and between the errors in $\bar{x}_{ct}$ and $\bar{y}_{ct}$ — estimated from the micro data within each cell
3	**Subtracting them removes the bias**	Consistent as the number of cohort-period observations grows, even with fixed cohort sizes

OLS on cohort means is the special case $\boldsymbol{\Sigma} = \mathbf{0}$, $\boldsymbol{\sigma} = \mathbf{0}$, i.e., pretending the cells are infinitely large.

Another Lens: The Pseudo-Panel as IV

Moffitt (1993): OLS on cohort-period means is equivalent to IV on the micro data, using **cohort** $\times$ **period dummies** as instruments for x_{ht}.

Why this helps

It turns “is my pseudo-panel any good?” into familiar IV questions: are the instruments *relevant*, and are they *exogenous*?

Relevance, in plain terms

Cohorts must **evolve differently** over time. If every cohort’s mean x moves in parallel, year effects absorb it and nothing is left to identify β.

Watch out for the age–period–cohort problem: with cohort and year fixed effects, a linear age effect is not separately identified, since age $=$ year $-$ birth year.

Choosing Cohort Size: A Tradeoff

Broad, few cohorts

Larger samples per cell shrink sampling error — but blur meaningful heterogeneity between members, and leave few panel units.

Narrow, many cohorts

Sharper, more homogeneous groups — but smaller cells mean noisier cohort means and bigger EIV corrections.

The extreme case

One all-inclusive “cohort” is the whole population: the model collapses into an aggregate macro time series.

Verbeek & Nijman (1992) find that with cells of a few hundred individuals the measurement-error bias in the FE estimator is often small enough to ignore; with small cells it is not.

A Bonus: Immunity to Attrition Bias

True individual panels

- — The same respondents, tracked year after year
- — Non-random dropout shrinks and skews the sample
- — Long panels are especially vulnerable
- — Attrition limits the useful length of the data

Pseudo-panels

- \+ A brand-new random sample in every survey
- \+ Representativeness refreshed by construction
- \+ Nobody to “drop out” of the cohort’s sampling frame
- \+ Can span very long periods

One exception: the cohort population itself can change through mortality or migration — a concern for older cohorts and long spans.

Pseudo-Panels: Strengths and Limitations

Strengths

+ Works when no true panel exists at all
+ Immune to attrition: fresh sample every survey
+ Can span very long time periods
+ Has a natural IV interpretation (Moffitt, 1993)

Limitations

— Needs large cells to control sampling error

— Needs an EIV correction, not plain OLS, when cells are small

— Loses within-cohort individual heterogeneity

— Differencing amplifies the measurement-error problem

— Identification needs cohorts that evolve differently

A Practical Checklist for Building a Pseudo-Panel

1	**Define cohorts with fixed membership**	Birth year, sex, birth region — traits that never change for an individual
2	**Check cell sizes**	Tabulate cohort $\times$ period counts; small cells mean large sampling error and a heavier EIV correction
3	**Keep the micro data**	You need it twice: for the means, and for their sampling variances
4	**Check that cohorts diverge**	Plot cohort means of x over time; parallel paths leave nothing to identify β
5	**Decide on levels vs. differences**	Differencing helps with non-stationarity but changes the error structure and amplifies noise

Pseudo-Panels in Stata (1/2): Benchmark First

```
clear all
webuse nlswork
xtset idcode year
xtreg ln_wage ttl_exp tenure wks_work i.year, fe vce(cluster idcode)
estimates store true_panel            // benchmark: the real panel FE

* Pretend we only have repeated cross-sections: one random year per woman
set seed 2026
generate double u = runiform()
bysort idcode (u): keep if _n == 1
```

`true_panel`	what we would estimate if we *could* follow each woman
`keep if _n == 1`	each woman appears once: independent cross-sections, checkable against the real panel

Pseudo-Panels in Stata (2/2): Build Cohorts and Estimate

```
generate byear = birth_yr
recode byear (41/43 = 43) (54 = 53)     // pool sparse extreme birth years
egen cohort = group(byear race)         // cohort = birth year x race
tabulate cohort year                    // check the cell sizes!
collapse (mean) ln_wage ttl_exp tenure wks_work ///
         (count) n = idcode, by(cohort year)    // one row per cohort-year
xtset cohort year
areg ln_wage ttl_exp tenure wks_work i.year [aweight = n], ///
     absorb(cohort) vce(cluster cohort)
estimates store pseudo_panel
estimates table true_panel pseudo_panel, b se keep(ttl_exp tenure wks_work)
```

`[aweight = n]` weights each cohort-year by its cell size; `absorb(cohort)` gives the cohort fixed effects θ_c^*.

In class: cells here are small and there are only $\sim$33 cohorts. How close is the pseudo-panel to the benchmark, and why might it differ?

Check Your Understanding

1	**Valid cohorts?**	Could you build cohorts by state of residence from a Brazilian household survey? What could go wrong?
2	**Cell size**	Your cohort-year cells have about 20 people each. Is OLS on cohort means safe? What do you do?
3	**Identification**	All cohorts' average education rises at the same pace. Can you estimate the return to education with cohort and year FE?
4	**Attrition**	Why is a pseudo-panel immune to attrition, and what kind of composition change can still bite?

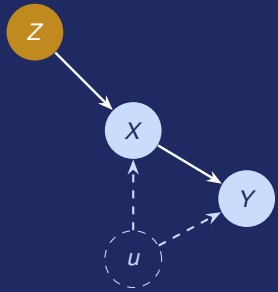
Recap and What's Next

Takeaway: pseudo-panels recover panel-like relationships — including fixed effects — from repeated cross-sections, at the cost of an errors-in-variables problem and the loss of individual-level detail.

Next: Hour 2 — IV Regressions in Panel Models

What to do when a regressor is still endogenous after fixed effects: combining the within transformation with two-stage least squares.

Questions?



INSTRUMENTAL VARIABLES

IV Regressions in Panel Models

Class 3 · Hour 2

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CLASS 3 · HOUR 2

What We'll Cover This Hour

01

Why endogeneity survives fixed effects

Time-varying confounders that FE alone cannot remove

02

2SLS mechanics, brought into the panel

The within transformation combined with two-stage least squares

03

Choosing instruments, internal or external

Lagged values, and the dynamic-panel complications they bring

04

Testing your instruments

Weak instruments, overidentification, and whether you needed IV at all

05

Estimating panel IV in Stata

Syntax, diagnostics, and the pitfalls that trip people up most

Recap: What We Have Seen

So far, fixed effects have handled confounders that are **constant** for each unit (and, with time FE, constant for each period).

> Today's question: even with fixed effects doing their job, what do you do when a regressor is **still correlated with** ε_{it}?

Fixed effects

Remove endogeneity that comes through α_i (and λ_t)

Instrumental variables

Remove endogeneity that comes through ε_{it}. Often you need both at once.

Endogeneity That Survives Fixed Effects

Fixed effects only remove what is constant for a unit. Anything that varies over time can still bias your estimate.

Simultaneity

y and *x* are jointly determined: investment and profitability move together within the same year

Time-varying omitted variables

A shock that changes both *x* and *y* in the same period, but was never measured

Measurement error

A time-varying regressor measured with noise — attenuation, which FE often makes worse

Dynamic feedback

Past outcomes shape current decisions, which shape future outcomes — violating strict exogeneity

The IV Idea, Refreshed

An instrument z isolates the part of x that is unrelated to the error, and uses only that part to estimate the effect on y.

Relevance

$$\text{Cov}(z_{it}, x_{it}) \neq 0 \quad \text{(after controls and FE)}$$

The instrument must actually predict the endogenous regressor. **Testable** in the first stage.

Exogeneity (exclusion)

$$\text{Cov}(z_{it}, \varepsilon_{it}) = 0$$

The instrument affects y only through x. **Never directly testable** — always an argument.

With fixed effects, "exogenous" means more: since demeaning mixes all periods, z must be uncorrelated with ε_{is} in *every* period s — strict exogeneity of the instrument.

The Panel IV Model

Structural equation, with x_{it} endogenous and $\boldsymbol{w}_{it}$ exogenous controls:

$$y_{it} = \beta_1 x_{it} + \boldsymbol{w}_{it}\boldsymbol{\delta} + \alpha_i + \varepsilon_{it}, \qquad \text{Cov}(x_{it}, \varepsilon_{it}) \neq 0$$

First stage, with the excluded instrument z_{it}:

$$x_{it} = \pi_1 z_{it} + \boldsymbol{w}_{it}\boldsymbol{\gamma} + \nu_i + \eta_{it}$$

The unit effect can live in *both* equations — which is exactly why FE and IV combine naturally: the within transformation removes α_i and ν_i at once.

Two-Stage Least Squares: The Mechanics

Stage 1

Regress the endogenous x_{it} on the instrument z_{it} and all exogenous controls. Keep the fitted values $\hat{x}_{it}$.

Stage 2

Regress y_{it} on $\hat{x}_{it}$ and the same controls. The coefficient on $\hat{x}_{it}$ is the 2SLS estimate.

Only $\hat{x}_{it}$ — the part of x explained by the instrument — identifies β_1. The rest of the variation in x is set aside.

Never run the two stages by hand. The point estimate is right, but the second-stage standard errors are wrong: they use residuals computed with $\hat{x}$ instead of x. Let the IV command do it.

Fixed Effects IV: Combining the Two

The within transformation and 2SLS are not in conflict: apply one, then the other.

1	**Demean everything first**	Subtract unit means from *y*, *x*, the controls *and* the instrument *z*, exactly as in ordinary FE
2	**Then run 2SLS on the demeaned data**	First stage: demeaned *x* on demeaned *z*; second stage uses the fitted values
3	**α_i cancels out algebraically**	So *z* may be correlated with α_i — it only needs to be (strictly) exogenous to ε_{it}
4	**One command does it all**	`xtivreg, fe` (official), or `xtivreg2` / `ivreghdfe` (user-written, with clustering and diagnostics)

The Alternative: Random Effects IV

RE-IV applies the same quasi-demeaning (by θ) as random effects, then runs 2SLS on the transformed data.

1	**Stronger requirement on the instruments**	z_{it} must be uncorrelated with *both* α_i and ε_{it} (the exogenous controls too)
2	**More efficient if that holds**	Uses between-unit variation in the instruments, not just within-unit variation
3	**Riskier if it doesn't**	The same fragility as plain RE, now compounded with the instrument's own assumptions

In practice: FE-IV is the common default. If something makes *x* endogenous with respect to ε_{it}, something probably makes it (and *z*) correlated with α_i too.

Choosing Instruments: External vs. Internal

External instruments

+ Come from outside the dataset: a policy change, a natural experiment, a geographic feature
+ Often the most credible, when available
− Hard to find one that is genuinely exogenous

Internal instruments

+ Built from the panel's own variables, e.g., lags of the endogenous regressor (see GMM)
+ Always available, no extra data needed
− Exogeneity is easy to violate — and hard to defend

With unit and year FE, an instrument that is constant within units, or common to all units in a year, is absorbed and becomes useless.

Internal Instruments: Lagged Values

The most common internal instrument: use $x_{i,t-1}$ (or a deeper lag) to instrument for x_{it}.

1	**Relevance is usually easy**	Most economic variables are persistent, so lags predict current values well
2	**Exogeneity needs no serial correlation**	If today's shock echoes into next period, $x_{i,t-1}$ is correlated with ε_{it}
3	**And with FE, it needs much more**	Demeaning brings future shocks into every period, so a lag that is only predetermined (not strictly exogenous) is invalid
4	**The usual fix: first differences**	Difference out α_i instead, and use lags dated $t-2$ or earlier: the logic of Arellano–Bond GMM

A Special Case: When x Is the Lagged y Itself

$$y_{it} = \rho\, y_{i,t-1} + \beta x_{it} + \alpha_i + \varepsilon_{it}$$

1. **Fixed effects alone are inconsistent** — Demeaning mechanically correlates $(y_{i,t-1} - \bar{y}_{i,-1})$ with $(\varepsilon_{it} - \bar{\varepsilon}_i)$: the Nickell (1981) bias
2. **Severe in short panels** — For large N, roughly $\text{plim}(\hat{\rho}^{FE} - \rho) \approx -\dfrac{1+\rho}{T-1}$: with $T = 10$ and $\rho = 0.5$, a bias of about -0.17
3. **The standard fix uses IV or GMM** — Arellano–Bond estimators instrument $\Delta y_{i,t-1}$ with $y_{i,t-2}$ and deeper lags

Dynamic GMM gets its own session later in the course. For now: know when to reach for it.

Weak Instruments: Why the First Stage Matters

A valid instrument that barely predicts x is nearly as bad as no instrument at all.

2SLS is biased toward OLS

With weak instruments, the 2SLS estimate drifts back toward the (endogenous) OLS or FE estimate

Inference breaks down

Conventional confidence intervals can be far too narrow, overstating what you know

Small invalidity, big bias

A slightly invalid instrument does much more damage when it is weak

Testing Instrument Relevance: The First-Stage F

How strongly do the excluded instruments predict x, after controls and fixed effects? The answer is a **size** of the F-statistic, not a p-value.

$F > 10$

The classic rule of thumb (Staiger & Stock, 1997)

Stock–Yogo

Formal critical values; e.g., 16.38 for 10% maximal size with one instrument

$F > 104.7$

For a valid 5% t-test with one instrument (Lee et al., 2022)

With clustered SEs, report the **Kleibergen–Paap rk Wald** F (shown by `ivreg2`, `xtivreg2`, `ivreghdfe`), or the effective F of Montiel Olea & Pflueger (2013). A first-stage $p < 0.05$ is *not* evidence of a strong instrument.

When You Have Extra Instruments: Overidentification

With more instruments than endogenous regressors, you can partially test the exogeneity assumption.

H_0: all instruments are valid (uncorrelated with ε_{it})

Sargan test (homoskedastic errors) or Hansen J test (robust / clustered) — use Hansen J in panels.

Reject H_0

At least one instrument is likely invalid — but the test can't say which one.

Fail to reject H_0

Consistent with validity: reassuring, not proof. If all instruments share the same flaw, the test cannot detect it.

Do You Even Need IV? The Durbin–Wu–Hausman Test

2SLS is less efficient than OLS or FE — so only pay that price if endogeneity is actually a problem.

H_0: x_{it} is exogenous ($\hat{\beta}^{FE}$ and $\hat{\beta}^{FE\text{-}IV}$ should be similar)

1	**Reject H_0**	The estimates differ meaningfully: endogeneity is real and IV is doing necessary work
2	**Fail to reject H_0**	No strong evidence of endogeneity: the simpler, more efficient FE estimate is defensible
3	**Only as good as the instrument**	The test assumes z is valid — and with a weak instrument it has little power

In Stata: the `endog()` option of `ivreg2`/`xtivreg2`/`ivreghdfe` gives a cluster-robust version.

Don’t Forget: Cluster the Standard Errors

Everything from Class 1, Hour 3 still applies: panel IV estimates need standard errors clustered by unit.

1	**Clustering doesn’t fix a bad instrument**	It corrects inference only; the estimate’s validity depends entirely on the instrument
2	**Both stages inherit the error structure**	First-stage *F*-statistics should be cluster-robust too (Kleibergen–Paap), not classical
3	**Few clusters need extra care**	With a handful of units (e.g., countries), consider the wild cluster bootstrap

Estimating Panel IV in Stata

```
clear all
webuse nlswork
xtset idcode year
* Official command (classical SEs):
xtivreg ln_wage not_smsa (tenure = union south) i.year, fe
* User-written: clustering + full IV diagnostics
ssc install ivreg2
ssc install ranktest
ssc install ftools
ssc install reghdfe
ssc install ivreghdfe
ivreghdfe ln_wage not_smsa (tenure = union south), absorb(idcode year) cluster(idcode) first
```

`(tenure = union south)`	`tenure` is endogenous; `union` and `south` are the excluded instruments
`absorb(idcode year)`	individual and year fixed effects
`cluster(idcode) first`	clustered SEs by woman, and the first-stage results

Reading the Output — and Questioning the Instruments

Kleibergen–Paap rk Wald *F*	Weak-instrument check (cluster-robust): compare with the thresholds on the first-stage slide
Hansen *J* statistic	Overidentification test: two instruments, one endogenous regressor, so one df
Underidentification test	Kleibergen–Paap rk LM: do the instruments identify the equation at all?

In class: this specification comes from Stata's own `xtivreg` examples, and is meant only to show syntax. Is `union` a credible instrument for `tenure`? Does union membership affect wages *only* through tenure?

Common Pitfalls in Panel IV

1	**Forgetting to demean the instrument**	Demeaning y and x but not z breaks the FE-IV logic entirely
2	**A "convenient" instrument without a story**	Statistical relevance isn't enough; you need an economic argument for exogeneity
3	**Ignoring a weak first stage**	A low F should change how much you trust the results, not just earn a footnote
4	**Running the two stages by hand**	Correct coefficients, wrong standard errors
5	**Instruments absorbed by the fixed effects**	Time-invariant z, or z common to all units in a year, has no within variation left

Check Your Understanding

1. **Which endogeneity?** You add firm FE to a regression of investment on cash flow. Name one source of endogeneity FE cannot remove.
2. **First stage** Your first-stage F is 6.2 with $p = 0.01$. Is the instrument strong? What do you report?
3. **Lagged instruments** Why is $x_{i,t-1}$ a questionable instrument for x_{it} in an FE-IV model?
4. **A state-level policy** A policy changes in all states in 2015 at once. Can it instrument x in a model with year FE?

Recap and What's Next

Takeaway: panel IV combines the within transformation with 2SLS to handle endogeneity that fixed effects alone can't reach — but it is only as credible as your instrument's exogeneity story and the strength of its first stage.

Next: Hour 3 — Paper Discussions

Research in practice. Our toolkit so far: pooled OLS, FE, RE, formal tests, time effects, pseudo-panels, and panel IV.

Questions?

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